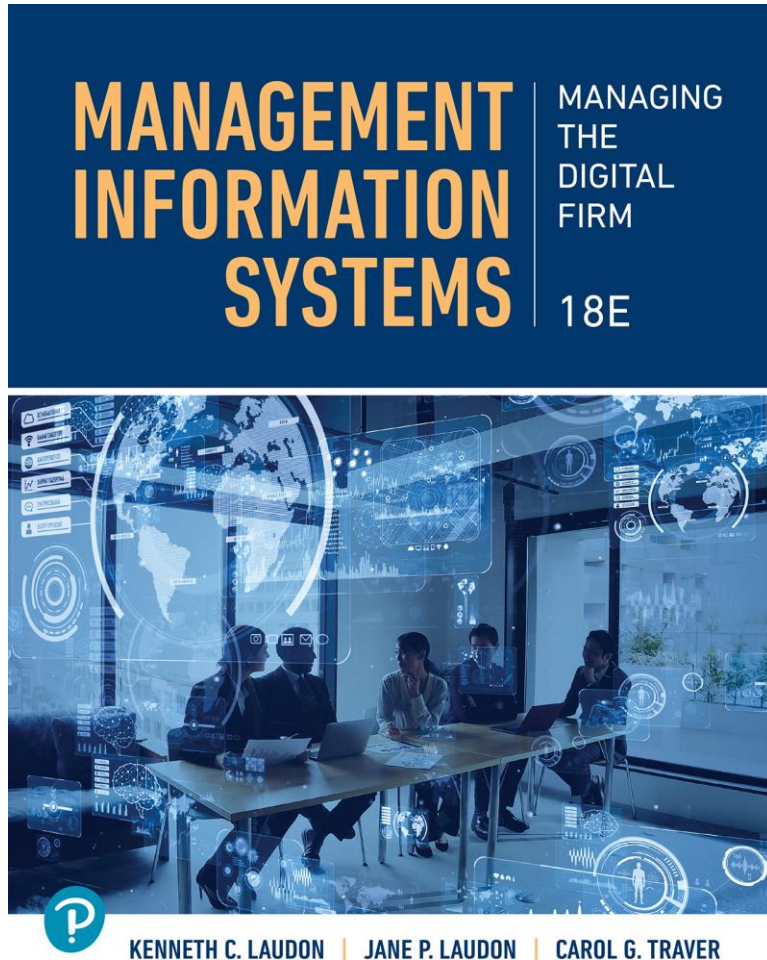


# Management Information Systems: Managing the Digital Firm

Eighteenth Edition



## Chapter 6

### Foundations of Business Intelligence: Databases and Information Management

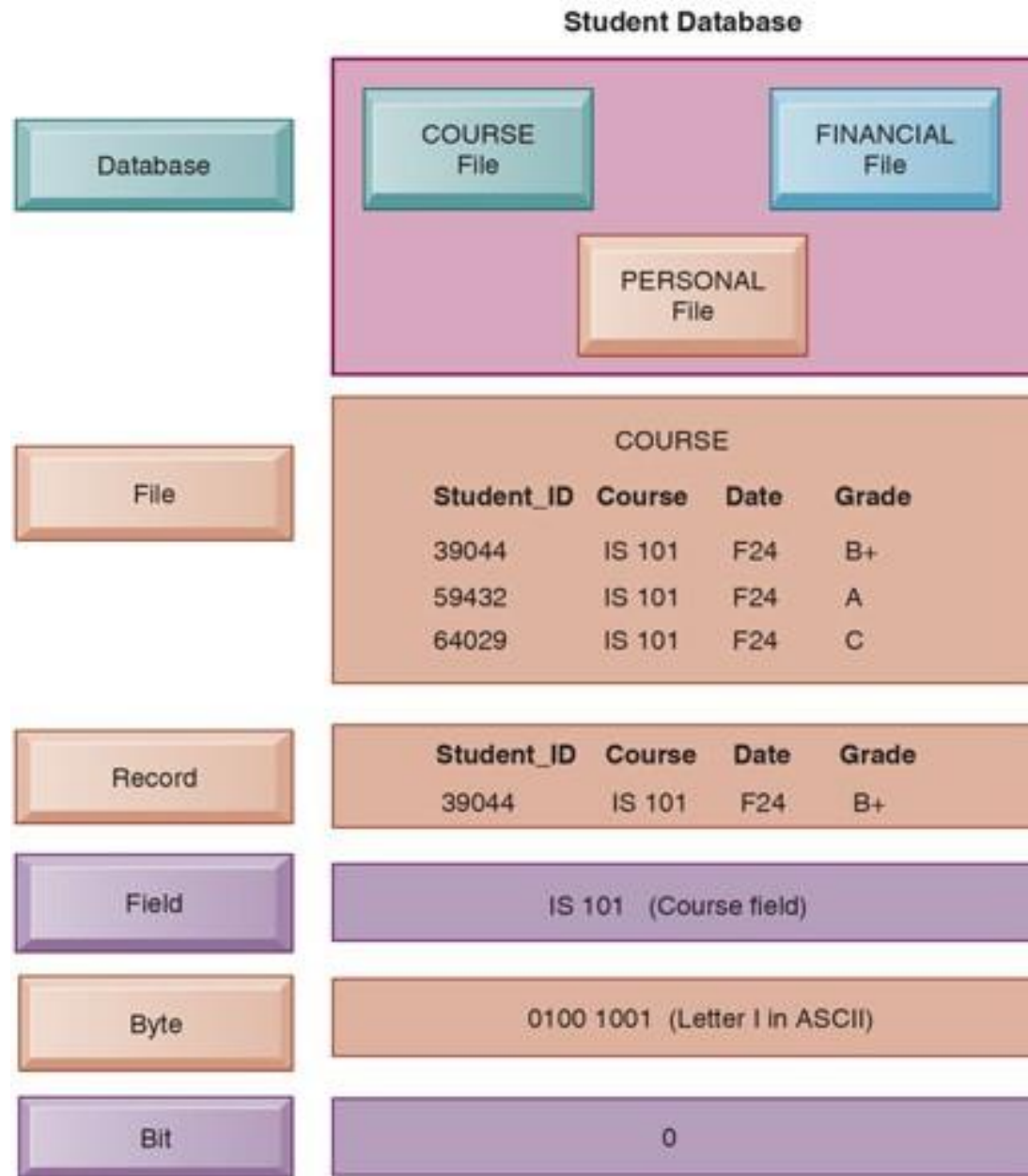
# Learning Objectives

- 6.1** Describe traditional file environment data management problems.
- 6.2** Describe DBMS and relational DBMS capabilities.
- 6.3** Discuss relational database design principles.
- 6.4** Describe non-relational, cloud, and blockchain database capabilities.
- 6.5** Describe tools used to analyze data in databases.
- 6.6** Discuss data governance, data quality assurance, data management.
- 6.7** Understand how the information in this chapter can help your career.

# Traditional File Environment's Data Management Problems (1 of 2)

- An effective information system provides users with accurate, timely, and relevant information
  - **Byte**: a single character of data
  - **Field**: Group of characters as word(s) or number(s)
  - **Record**: Group of related fields
  - **File**: Group of records of same type
  - **Database**: Group of related files
  - **Entity**: Person, place, thing on which we store information
  - **Attribute**: Each characteristic, or quality, describing entity

# Figure 6.1 The Data Hierarchy

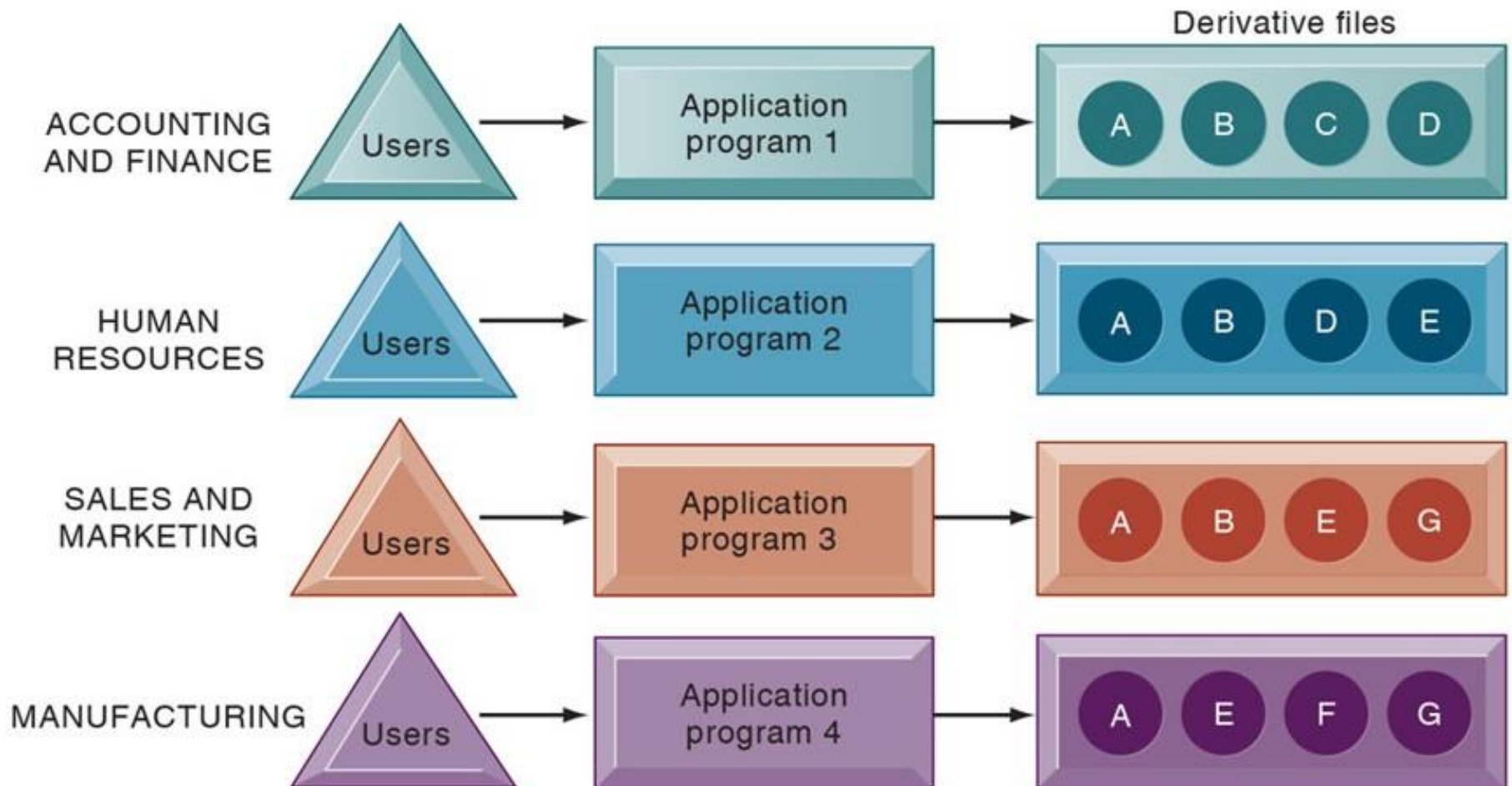


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# Traditional File Environment's Data Management Problems (2 of 2)

- Problems with the traditional file environment
  - Files maintained separately by different departments
  - Data redundancy
  - Data inconsistency
  - Program-data dependence
  - Lack of flexibility
  - Poor data security
  - Inability to share data among applications

# Figure 6.2 Traditional File Processing



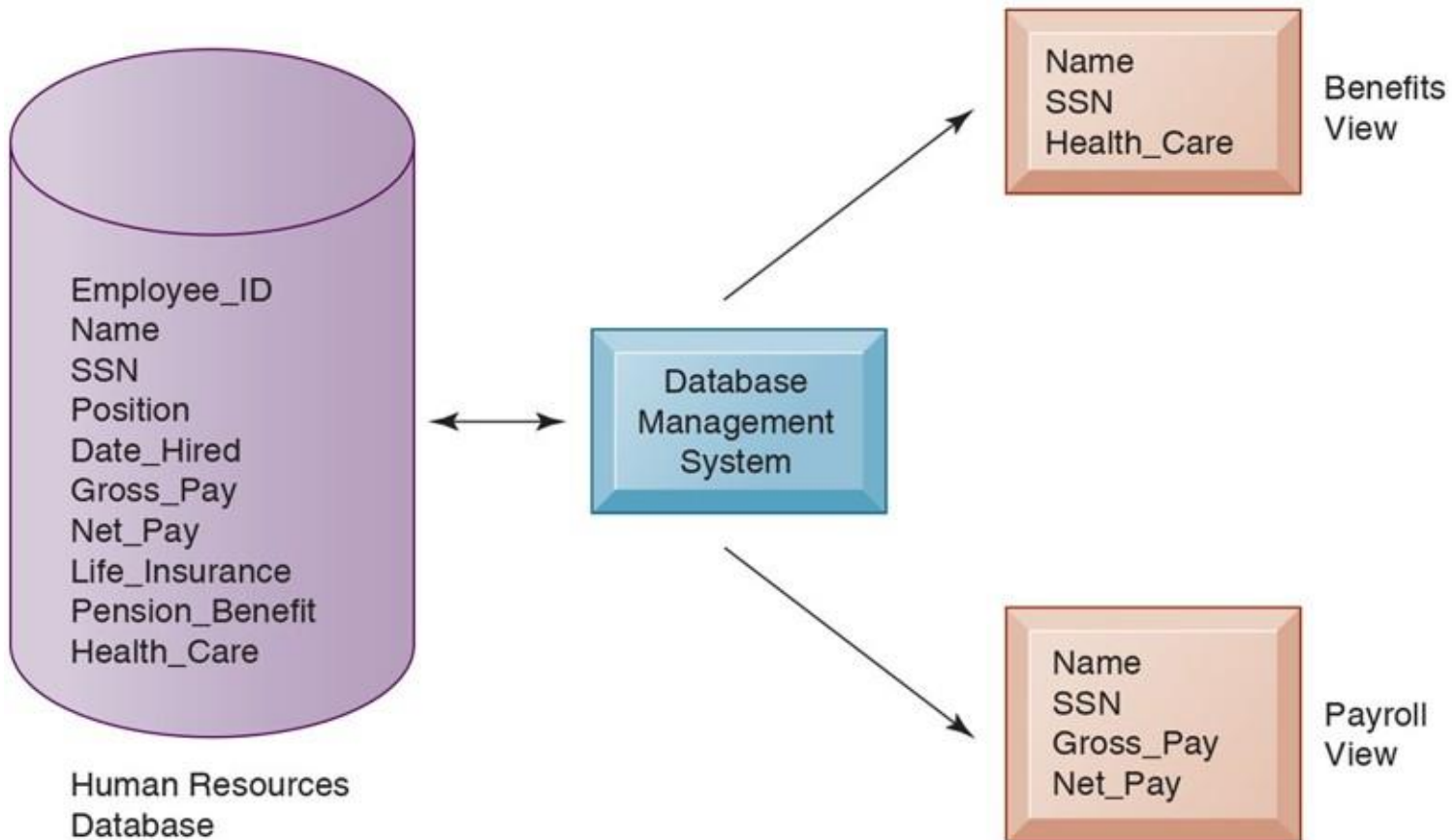
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# DBMS and Relational DBMS Capabilities

(1 of 4)

- A database management system (DBMS)
  - Software that enables an organization to
    - Centralize data
    - Manage that data efficiently, and
    - Provide access to the stored data by application programs
  - A DBMS reduces data redundancy and inconsistency by minimizing isolated files in which the same data are repeated

# Figure 6.3 Human Resources Database with Multiple Views



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# DBMS and Relational DBMS Capabilities

(2 of 4)

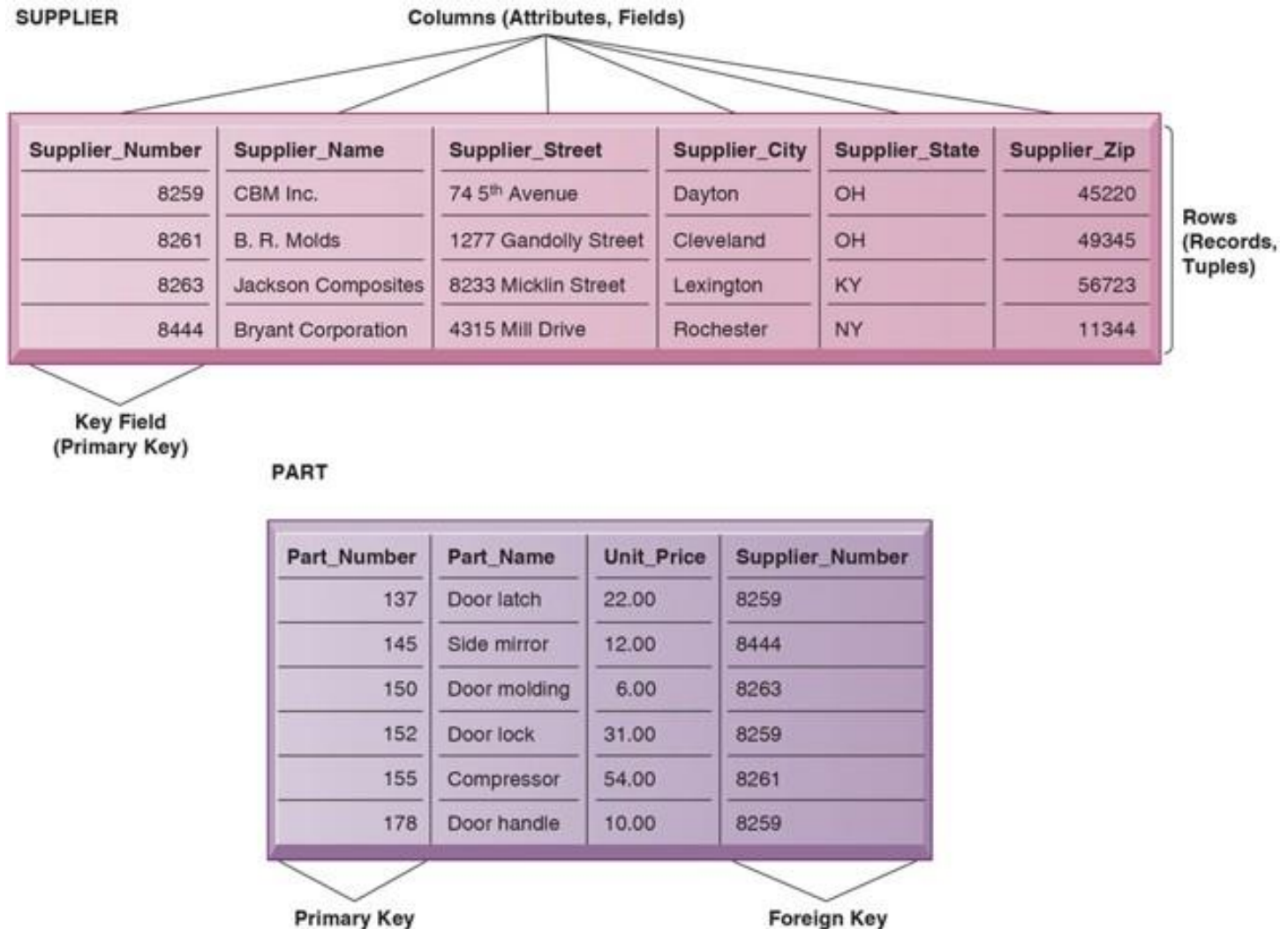
- Contemporary DBMS use different database models to keep track of entities, attributes, and relationships
- Relational database
  - Most common type of database
  - Represents data as two-dimensional tables (called relations)
  - Each table contains data on entity and attributes
- Relational DBMS (RDBMS)
  - Most popular type of DBMS today for desktops as well as for larger computers and mainframes

# DBMS and Relational DBMS Capabilities

(3 of 4)

- A table is a grid of columns and rows
- The structure of a database includes
  - **Rows:** Records for different entities
  - **Fields (columns):** Represents attribute for entity
  - **Key field:** Field used to uniquely identify each record
  - **Primary key:** Field in table used for key fields
  - **Foreign key:** Primary key used in second table as look-up field to identify records from original table

# Figure 6.4 Relational Database Tables



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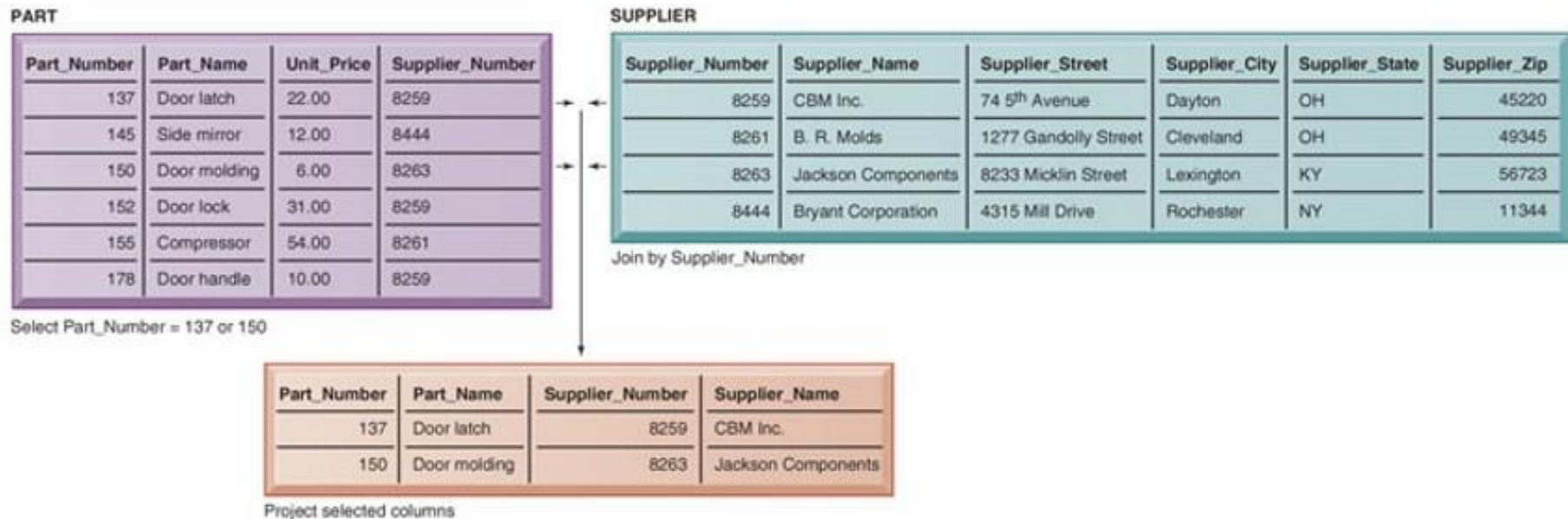
# DBMS and Relational DBMS Capabilities

(4 of 4)

- Three basic operations of a relational DBMS
  - SELECT
    - Creates subset of data of all records that meet stated criteria
  - JOIN
    - Combines relational tables to provide user with more information than available in individual tables
  - PROJECT
    - Creates subset of columns in table, creating tables with only the information specified

# Figure 6.5

## Three Basic Operations of a Relational DBMS



# Capabilities of DBMS

- Data definition
- Data dictionary
- Querying and reporting
  - Data manipulation language
  - Structured Query Language (SQL)
- Many DBMS have report generation capabilities for creating polished reports (Microsoft Access)

# Figure 6.6 Microsoft Access Data Dictionary Features

**Navigation Pane**

**Table Design**

File Home Create External Data Database Tools Help **Table Design** Tell me what you want to do

Views Primary Key Builder Test Validation Rules Insert Rows Delete Rows Modify Lookups Property Indexes Sheet Show/Hide Create Data Macros Rename/Delete Macro Relationships Object Dependencies Relationships

**SUPPLIER**

| Field Name      | Data Type  | Description (Optional)         |
|-----------------|------------|--------------------------------|
| Supplier_Number | Number     | Supplier Identification Number |
| Supplier_Name   | Short Text | Supplier Name                  |
| Supplier_Street | Short Text | Supplier Street                |
| Supplier_City   | Short Text | Supplier City                  |
| Supplier_State  | Short Text | Supplier State                 |
| Supplier_Zip    | Short Text | Supplier Zip                   |

**Field Properties**

General Lookup

|                 |                     |
|-----------------|---------------------|
| Field Size      | Long Integer        |
| Format          |                     |
| Decimal Places  | Auto                |
| Input Mask      |                     |
| Caption         |                     |
| Default Value   |                     |
| Validation Rule |                     |
| Validation Text |                     |
| Required        | Yes                 |
| Indexed         | Yes (No Duplicates) |
| Text Align      | General             |

Design view. F6 = Switch panes. F1 = Help.

A field name can be up to 64 characters long, including spaces. Press F1 for help on field names.

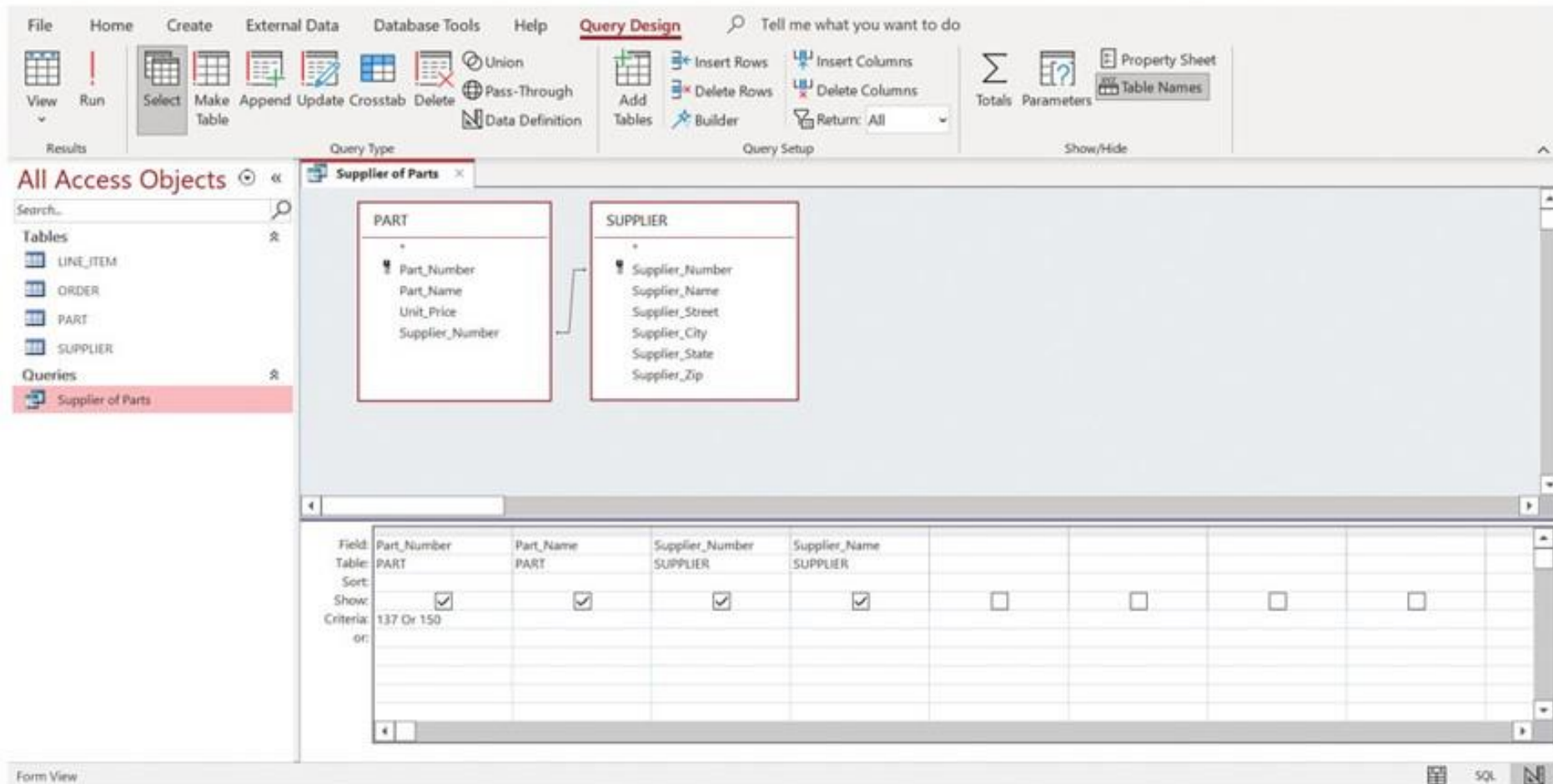
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## Figure 6.7 Example of an SQL Query

```
SELECT PART.Part_Number, PART.Part_Name, SUPPLIER.Supplier_Number,  
SUPPLIER.Supplier_Name  
FROM PART, SUPPLIER  
WHERE PART.Supplier_Number = SUPPLIER.Supplier_Number AND  
Part_Number = 137 OR Part_Number = 150;
```

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# Figure 6.8 Microsoft Access Query



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# Relational Database Design Principles

(1 of 2)

- Conceptual design vs. physical design
- Normalization
  - Streamlining complex groupings of data to minimize redundant data elements and awkward many-to-many relationships

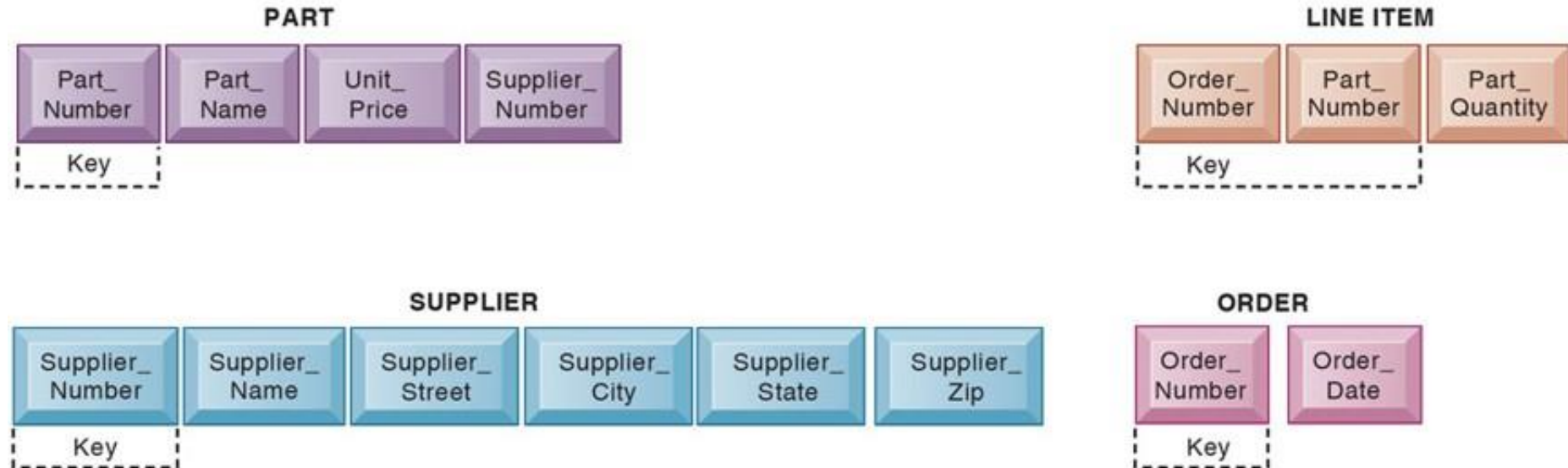
# Figure 6.9 Unnormalized Relation for ORDER

**ORDER (Before Normalization)**

|                  |                |                 |               |                |                   |                     |                   |                     |                   |                    |                  |
|------------------|----------------|-----------------|---------------|----------------|-------------------|---------------------|-------------------|---------------------|-------------------|--------------------|------------------|
| Order_<br>Number | Order_<br>Date | Part_<br>Number | Part_<br>Name | Unit_<br>Price | Part_<br>Quantity | Supplier_<br>Number | Supplier_<br>Name | Supplier_<br>Street | Supplier_<br>City | Supplier_<br>State | Supplier_<br>Zip |
|------------------|----------------|-----------------|---------------|----------------|-------------------|---------------------|-------------------|---------------------|-------------------|--------------------|------------------|

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# Figure 6.10 Normalized Tables Created from ORDER



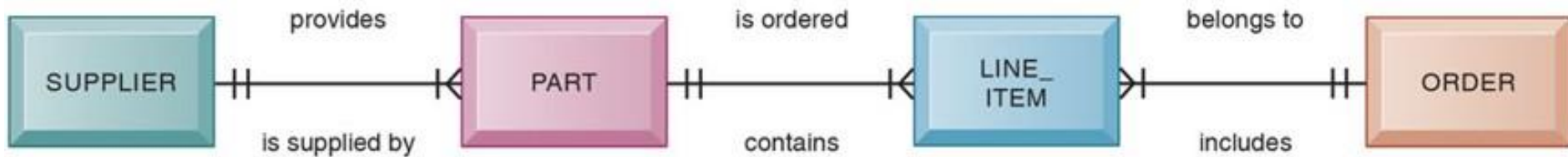
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# Relational Database Design Principles

## (2 of 2)

- Referential integrity
  - Rules used by RDBMS to ensure relationships between tables remain consistent
- Entity-relationship diagram
  - Illustrates the relationship among various entities in the database

# Figure 6.11 An Entity-Relationship Diagram



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# Non-Relational Database, Cloud Database, and Blockchain Capabilities (1 of 3)

- Non-relational database management system:  
**“No SQL”**
  - More flexible data model
  - Data sets stored across distributed machines
  - Easier to scale
  - Handle large volumes of unstructured & structured data

# Non-Relational Database, Cloud Database, and Blockchain Capabilities (2 of 3)

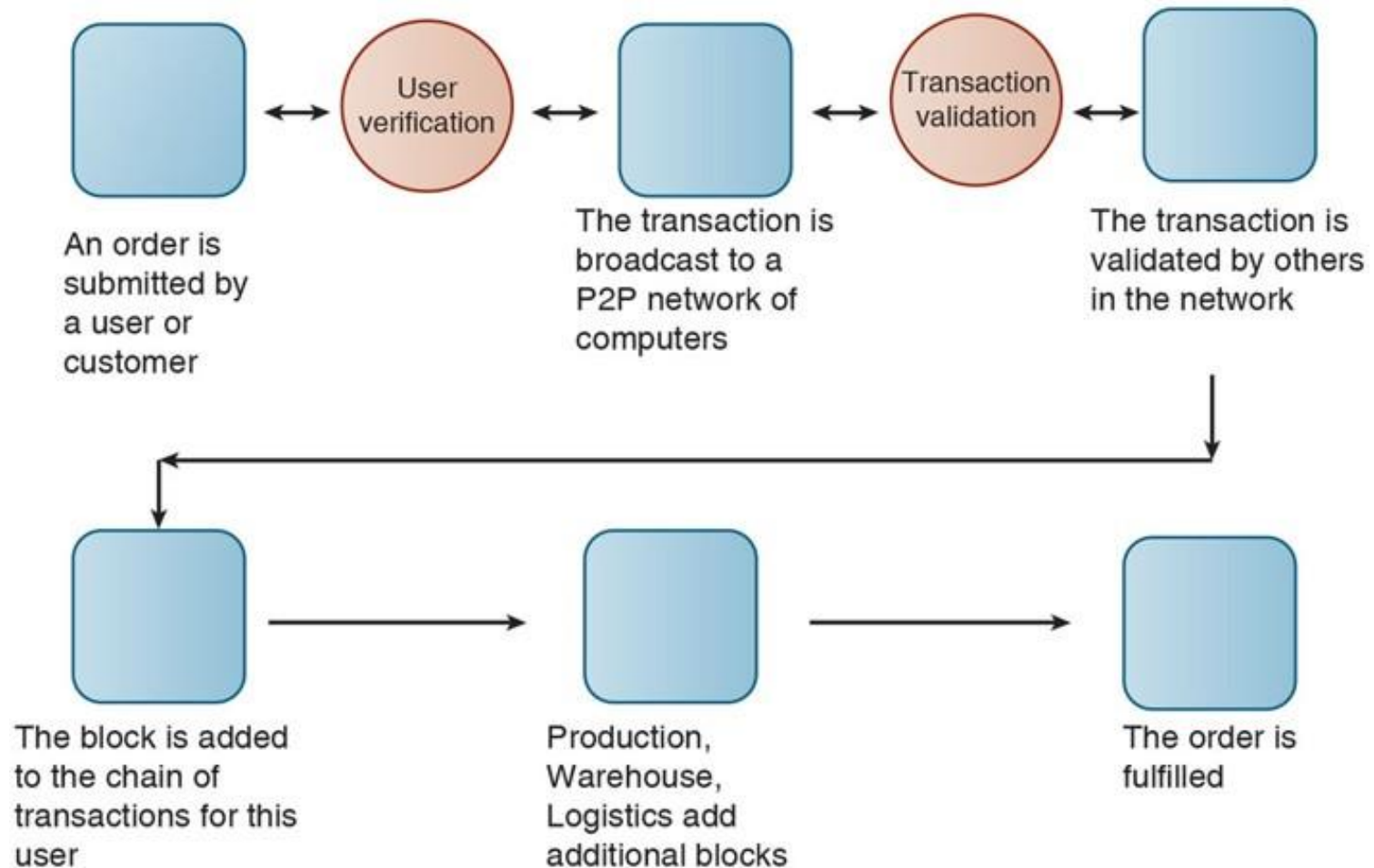
- Cloud databases
  - Appeal to start-ups, smaller businesses
  - Amazon Relational Database Service, Microsoft SQL Azure
  - Private clouds
- Distributed databases
  - Stored in multiple physical locations
  - Example: Google Spanner

# Non-Relational Database, Cloud Database, and Blockchain Capabilities (3 of 3)

- Blockchain

- Distributed ledgers in a peer-to-peer distributed database
- Maintains a growing list of records and transactions shared by all
- Encryption used to identify participants and transactions
- Used for financial transactions, supply chain, and medical records
- Foundation of Bitcoin, and other crypto currencies

# Figure 6.12 How Blockchain Works



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## Examples of Blockchain Applications

| Business Specialty      | Organization          | Blockchain Application                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Supply chain management | IBM Food Trust        | IBM Food Trust, a collaborative network of growers, distributors, manufacturers, and retailers, is using blockchain to track and trace goods through the supply chain, from raw materials to finished products. The solution connects participants through a permissioned, immutable, and shared record of food provenance, transaction data, processing details and more. This can help to ensure transparency and traceability, and to reduce fraud. |
| Healthcare              | Government of Estonia | The government of Estonia is using blockchain to create secure and tamper-proof electronic health records (EHRs). This can help to improve the quality of care and to reduce fraud.                                                                                                                                                                                                                                                                    |
| Banking                 | Ripple                | Blockchain can be used to make cross-border payments faster and cheaper, as it eliminates the need for intermediaries such as banks. For example, the Ripple blockchain-based payment network allows users to send and receive money in different currencies quickly and easily.                                                                                                                                                                       |
| Cryptocurrency          | Circle                | Circle is a fintech platform that oversees the exchange of traditional and cryptocurrency payments between users. Circle's merchant payment services move money between digital currencies securely and quickly. Circle accepts crypto payments in currencies like Bitcoin, Ethereum, USD Coin and Euro Coin.                                                                                                                                          |
| Shipping and Logistics  | DHL                   | DHL uses blockchain to keep a digital ledger of shipments and maintain integrity of transactions.                                                                                                                                                                                                                                                                                                                                                      |

# Tools Used to Analyze Data in Databases

- Big data
  - Massive sets of unstructured/semi-structured data from web traffic, social media, sensors, and so on
  - Volumes too great for typical DBMS
    - Petabytes, exabytes of data
- Big data and AI have a synergistic relationship
  - AI requires a massive amount of data for training
  - Machine learning is able to use the information provided by big data to generate more valuable business insights

## Table 6.2

### Big Data and AI Applications

| Organization    | Big Data and AI Applications                                                                                                                                                                                                                                            |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| McDonalds       | Uses big data and AI to analyze drive-thru traffic patterns and identify ways to improve efficiency such as determining the optimal number of order lanes and the best way to staff them. Also uses AI in voice recognition systems that can help to reduce wait times. |
| JP Morgan Chase | Uses big data to assess creditworthiness, detect fraud, and manage risk. The company also uses AI to develop algorithms for trading and investment.                                                                                                                     |
| Google Health   | Uses big data and AI to accurately interpret retinal scans to detect diabetic retinopathy. A large team of ophthalmologists helped Google review more than 100,000 retinal scans to develop an AI-based application called Automated Retinal Disease Assessment.        |
| Siemens         | Advanced Artificial Intelligence Services use big data and AI to monitor factory equipment, predict downtime, and optimize production processes. Also uses AI to develop robots.                                                                                        |
| TripAdvisor     | Uses AI-based sentiment analysis to process the feedback received from millions of travelers. This method enables TripAdvisor to identify essential insights within customer reviews, personalize recommendations, and enhance its services.                            |

# Business Intelligence Infrastructure

(1 of 4)

- Array of tools for obtaining information from separate systems and from big data
  - Data warehouse
  - Data mart
  - Data lake
  - Hadoop
  - In-memory computing
  - Data analytics platform

# Business Intelligence Infrastructure

(2 of 4)

- **Data warehouse**

- Database that stores current and historical data from many core operational transaction systems
- Consolidates and standardizes information for use across enterprise, but data cannot be altered
- Provides analysis and reporting tools

- **Data mart**

- Subset of data warehouse
- Typically focus on single subject or line of business

# Business Intelligence Infrastructure

(3 of 4)

- **Data lake**

- Repository for raw unstructured data or structured data that, for the most part, has not yet been analyzed

- **Hadoop**

- Open-source software framework managed by the Apache Software Foundation
- Enables distributed parallel processing of big data across inexpensive computers

# Business Intelligence Infrastructure

(4 of 4)

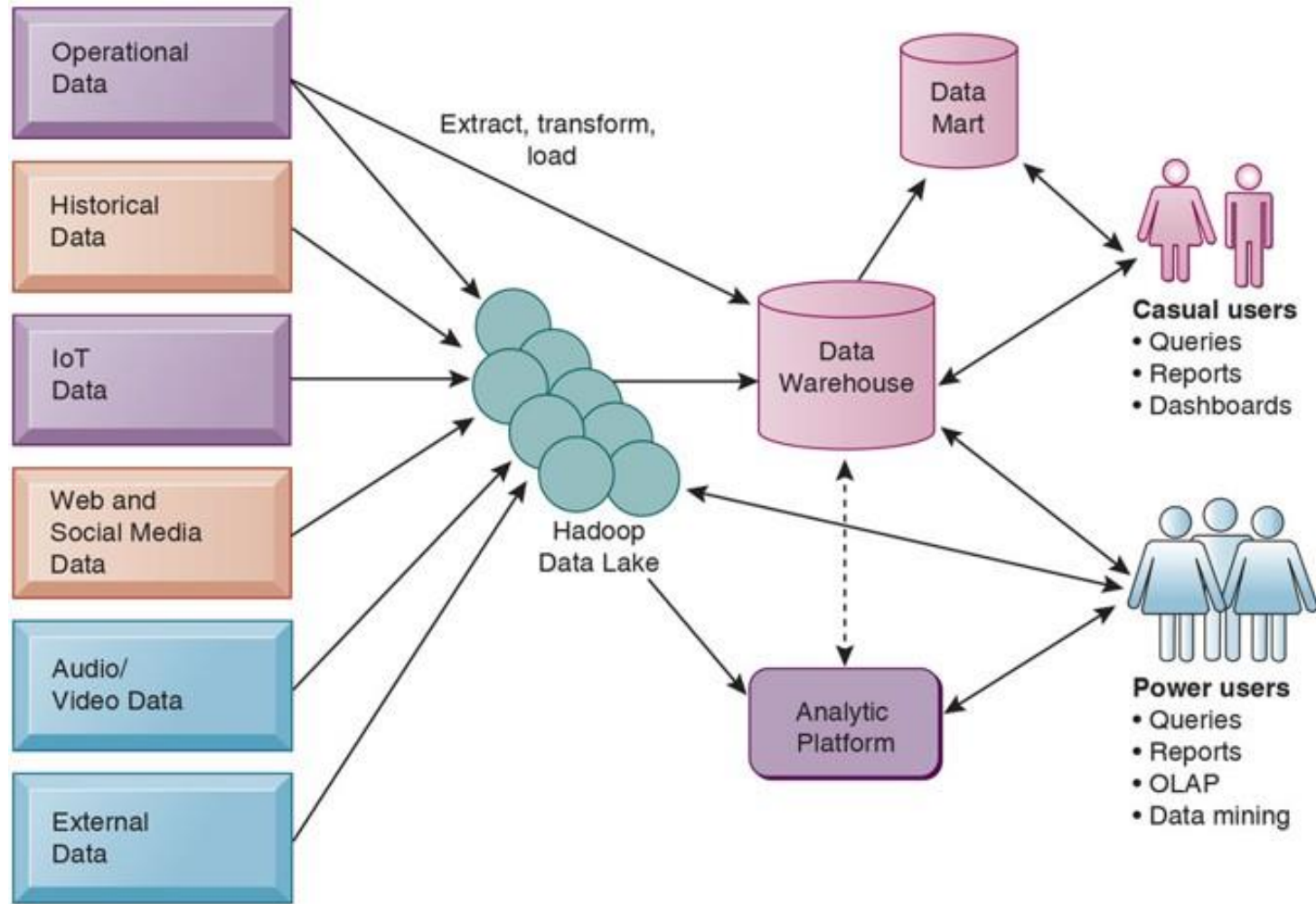
- **In-memory computing**

- Uses computers main memory (RAM) for data storage to avoid delays in retrieving data from disk storage
- Can reduce hours/days of processing to seconds
- Requires optimized hardware

- **Data-analytic platform**

- High-speed platform using both relational and non-relational tools optimized for large datasets

# Figure 6.13 Contemporary Business Intelligence Technology Infrastructure



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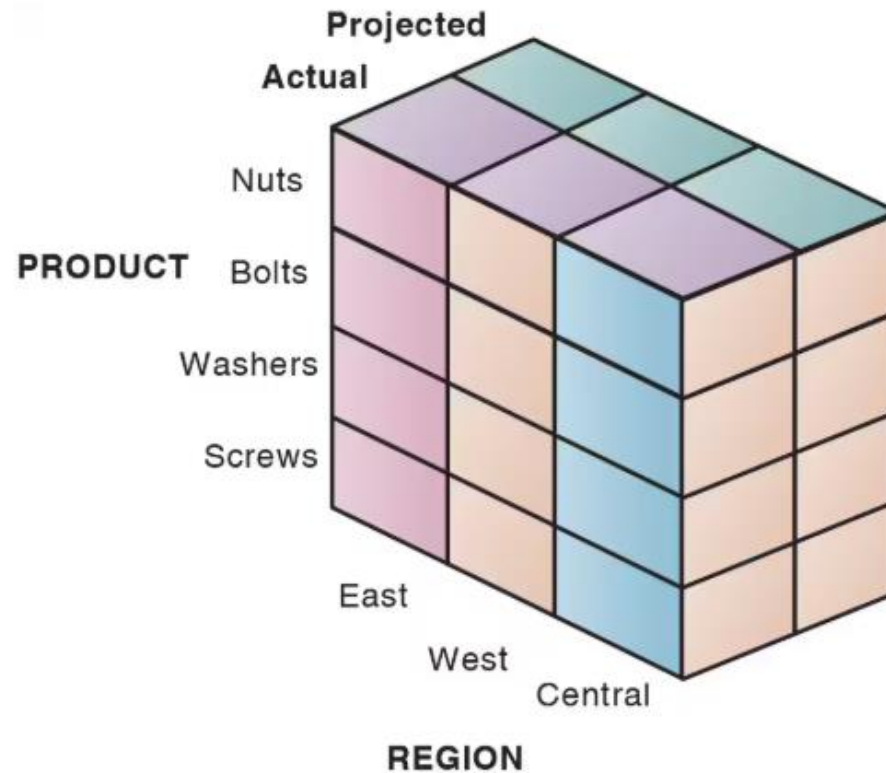
# Analytical Tools: Relationships, Patterns, Trends (1 of 2)

- Tools for consolidating, analyzing, and providing access to vast amounts of data to help users make better business decisions
  - **Online analytical processing (OLAP)**
    - Supports multidimensional data analysis, enabling users to view the same data in different ways using multiple dimensions
  - **Data mining**
    - Provides insights into corporate data that cannot be obtained with OLAP
      - [Associations](#) are occurrences linked to a single event while in [sequences](#), events are linked over time.
      - [Classification](#) recognizes patterns that describe the group to which an item belongs by examining existing items that have been already classified. [Clustering](#) works in a manner similar to classification when no groups have yet been defined.
      - [Forecasting](#) uses a series of existing values to forecast what other values will be.

# Example Data Model for OLAP

**Figure 6.14**

Multidimensional Data Model



This view shows actual sales of each product in each region. If other views are possible.

# Analytical Tools: Relationships, Patterns, Trends (2 of 2)

- Tools for consolidating, analyzing, and providing access to vast amounts of data to help users make better business decisions

## —Text mining

- Tools to extract key elements from unstructured natural language text, discover patterns and relationships, and summarize the information
- **Sentiment analysis**
  - Involves mining comments in an email, etc. to detect favorable and unfavorable opinions about specific subjects

## —Web mining

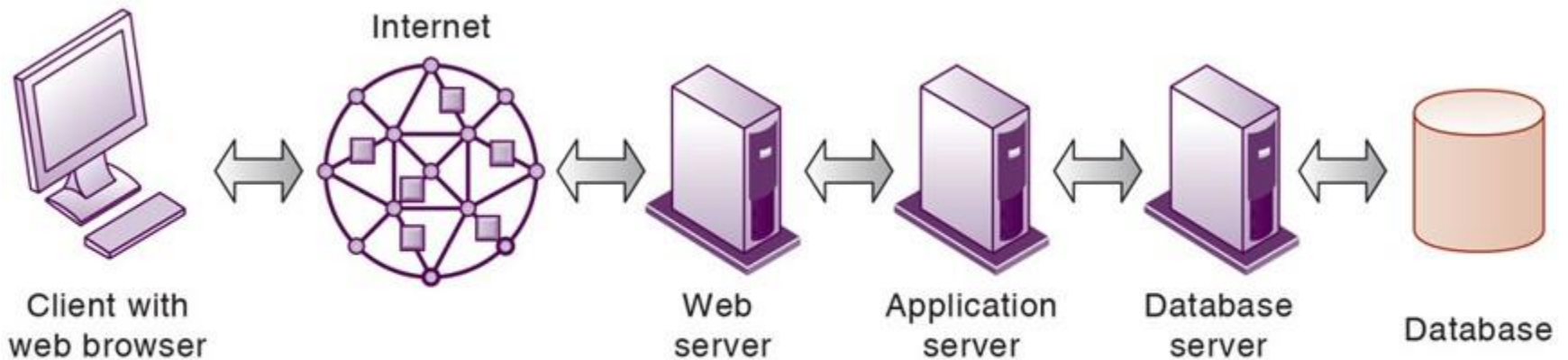
- Discovery and analysis of useful patterns and information from the web
  - E.g., Hotjar Heatmaps to identify which elements visitors clicked on the most and least and how far down the page they scrolled
    - <https://www.hotjar.com/customers/techsmith/>

# Databases and the Web

- Many companies use the web to make some internal databases available to customers or partners
- Typical configuration includes
  - Web server
  - Application server/middleware/scripts
  - Database server (hosting DBMS)
- Advantages of using the web for database access
  - Ease of use of browser software
  - Web interface requires few or no changes to database
  - Inexpensive to add web interface to system

# Figure 6.15

## Linking Internal Databases to the Web



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# Data Governance, Data Quality Assurance, and Data Management (1 of 3)

- **Data governance**

- Encompasses policies and procedures through which data can be managed as an organizational resource.
- Establishes rules for sharing, disseminating, acquiring, standardizing, classifying and inventorying information
  - Example: Firm information policy that specifies that only selected members of a particular department can view certain information

# Data Governance, Data Quality Assurance, and Data Management (2 of 3)

- **Data quality**

- More than 25 percent of critical data in Fortune 1000 company databases are inaccurate or incomplete
- Before new database is in place, a firm must:
  - Identify and correct faulty data
  - Establish better routines for editing data once database in operation

# Data Governance, Data Quality Assurance, and Data Management (3 of 3)

- **Data quality audit**

- Analysis of data quality often begins with a data quality audit
  - A structured survey of the accuracy and level of completeness of the data in an information system

- **Data cleansing**

- Also known as data scrubbing
- Consists of processes for detecting and correcting data in a database or file that are incorrect, incomplete, improperly formatted, or redundant

# How AI Improves Data Management

- Data management is crucial for creating an environment where data can be shared and utilized across the entire organization
  - However, managing all of the organization's data is extremely labor-intensive
  - Artificial intelligence is now being used to improve data management in the following areas:
    - classification, cataloguing, improving data quality, security, data integration, and data analysis

| Problem/Challenge      | Description                                                                                                                                                         | AI Solution                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Classification         | Obtaining, extracting, and structuring data from documents, photos, handwriting, video, and other media into categories.                                            | AI helps identify inappropriate social media postings (such as hate speech) on a large scale, including postings generated by bots. Also used for tasks such as identifying the most important data in documents such as a contract.                                                                                                                                                        |
| Cataloguing            | Locating data and keeping the data current. Extremely labor-intensive.                                                                                              | AI can automate searches through various repositories of data and create catalogs automatically. Can also identify where data originated, how it has been modified, and where it resides.                                                                                                                                                                                                   |
| Data Quality Assurance | Reducing errors in data                                                                                                                                             | AI machine learning tools can scan data to identify values that are not allowed and can correct some errant values automatically. AI algorithms can detect missing values in the data and then can provide more accurate estimated values.                                                                                                                                                  |
| Security               | Ensuring that data can't be stolen, altered, or destroyed by unauthorized parties and that data usage conforms to organizational policies.                          | AI machine learning solutions automate collection of security data across multiple internal and external systems and assess which threats are most credible. AI systems can predict likely attack paths based on previous attack patterns and determine whether new threats are coming from previously known actors or new ones. AI can also be used to identify internal threats of fraud. |
| Data Integration       | Consolidating data from different sources into a unified view [single cohesive dataset] that can be analyzed for business intelligence.                             | Machine learning-based mastering systems use probabilistic matching techniques to decide whether records should be combined. Records that have a high probability of being the same entity are automatically merged. Records that can't be resolved by this approach can be reviewed by human subject matter experts.                                                                       |
| Data Analysis          | Ability to draw conclusions from or evaluate specific data. Data analysis is used to make purposeful discoveries, suggest conclusions, and support decision-making. | <p>AI and machine learning can process vast amounts of data quickly-and often more accurately than humans. AI can identify complex patterns and provide predictive insights that may not be readily apparent through manual analysis.</p> <p>Natural language processing tools, including ChatGPT, can analyze textual data, while computer vision systems can analyze images.</p>          |

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